Heart Disease Prediction: Using LR & ANN

Team Members

Karun Lamsal	THA081BCT013
Kishan Kumar Shah	THA081BCT014

Under the Supervision of:

Er. Suwarna Lingden

Department of Electronics and Computer Engineering

Institute of Engineering, Thapathali Campus

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Presentation Outline

- Introduction
- Math Formulation
- Dataset & Cleaning
- EDA
- Feature Engineering
- Model Development
- Evaluation
- Results & Conclusion

Introduction

- **Goal**: Predict heart disease (yes/no) from 13 clinical features
- **Dataset**: UCI Heart Disease - 302 records
- **Models**: Logistic Regression vs ANN
- **Why**?
  CVDs kill 17.9M people/year (WHO)
  - early detection saves lives

Mathematical Formulation[1]

- **Logistic Regression**

$$\boldsymbol{z = w_{1x_1} + w_{2x_2} + \ldots + w_{nx_n} + b}$$

$$\boldsymbol{\sigma(z) = \frac{1}{1 + e^{\{-z\}}}}$$

$$\boldsymbol{L = -\frac{1}{n}\sum[y_i \log(p_i) + (1 - y_i)\log(1 - p_i)]}$$

- **LR**: weighted **sum** of features **-- sigmoid** squishes **to probability [0, 1]** –

  above **threshold**(0.5) = disease

- **Loss function** (**binary cross-entropy**) measures **prediction error**; model minimizes it during training

Mathematical Formulation[2]

- **ANN**

$$\boldsymbol{output} = \mathrm{ReLU}(\boldsymbol{w_1} \cdot \boldsymbol{x_1} + \boldsymbol{w_2} \cdot \boldsymbol{x_2} + \cdots + \boldsymbol{w_{13}} \cdot \boldsymbol{x_{13}} + \boldsymbol{b})$$

$$\boldsymbol{f(x)} = \mathbf{max}(\mathbf{0}, \boldsymbol{x})\ (\mathrm{ReLU})$$

$$\boldsymbol{w_{new}} = \boldsymbol{w_{old}} - \boldsymbol{\alpha} \cdot \frac{\boldsymbol{\partial L}}{\boldsymbol{\partial w}}$$

- **ANN**: **ReLU** in hidden layers (**zeros out negatives**) -- weights updated via gradient descent
- **Gradient** shows each weight's error contribution -- weight adjusted accordingly

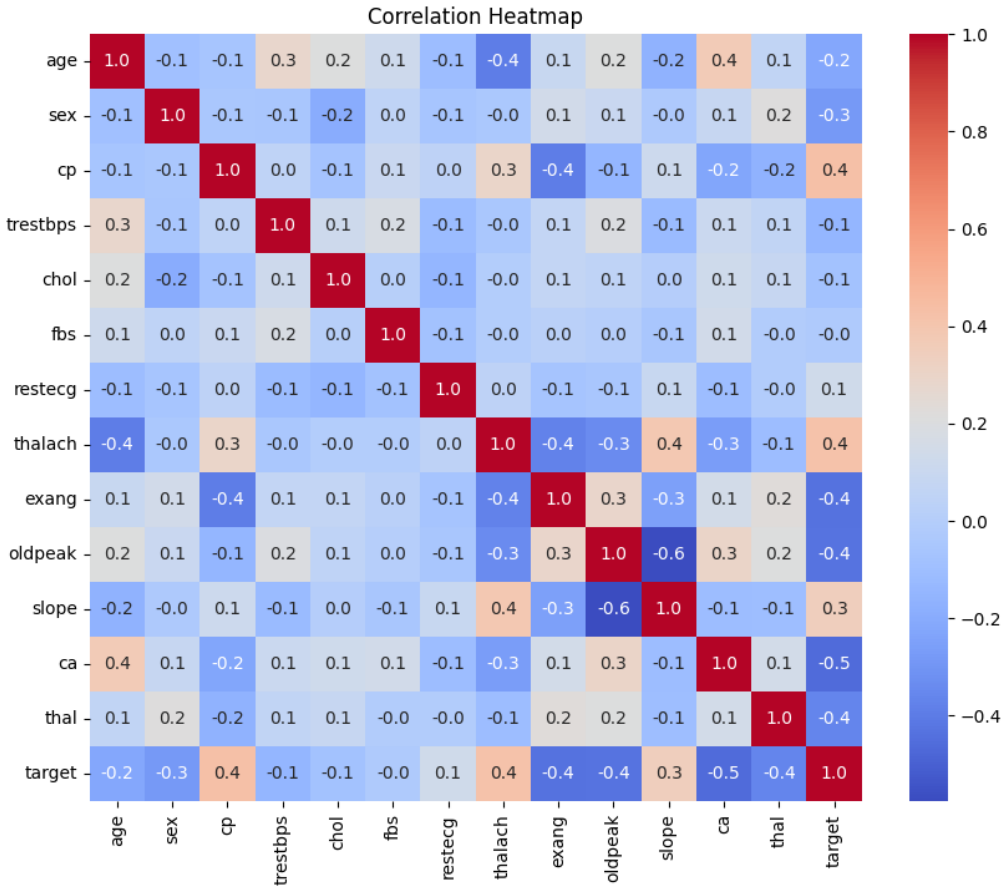
Dataset Description & Data Cleaning

- UCI Heart Disease Dataset - **302 × 14**

Feature	**Type**	**Feature**	**Type**
age	Continuous	exang	Binary
sex	Binary	oldpeak	Continuous
cp	Categorical	slope	Categorical
trestbps	Continuous	ca	Ordinal
chol	Continuous	thal	Categorical
fbs	Binary	**target**	**Binary**
restecg	Categorical	thalach	Continuous

- No **NaN** values, no **duplicates**
- **Cleaning**: thal=0 (2 rows) and ca=4 (4 rows) - invalid, replaced with median
- **Final**: 302 samples ready

Exploratory Data Analysis



- **Target**: 164 disease (**54%**) / 138 healthy (**46%**) – balanced
- **Top correlations**: cp (+0.4), thalach (+0.4), ca (−0.4), exang (−0.4)

Feature Engineering & Preprocessing

- **X** (**features**/input var) & **y** (**target**/output var) >> **predict y from X**.
- **Train**: **241** (80%) | **Test**: **61** (20%) - **stratified** split >> same disease ratio in both sets (unbiased)
- **StandardScaler**: $z = \frac{x-\mu}{\sigma}$
- Fit (learn stats) on training data only; transform (scale) test data with same parameters to prevent data leakage.
- Needed because **chol** is in **hundreds**, **sex** is **0/1** -> without scaling, large features **dominate**

Model Development [LR] [1]

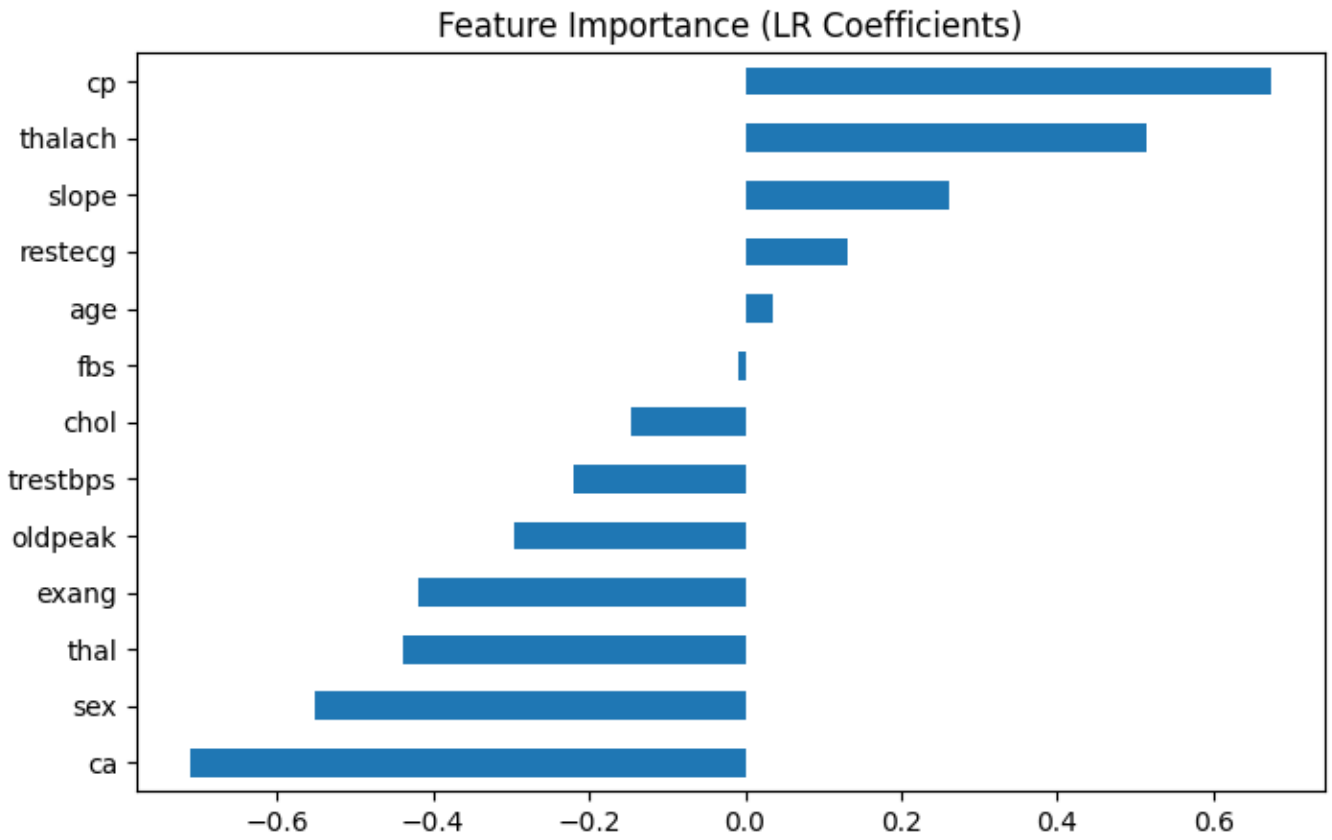
- **GridSearchCV (5-fold CV)** : tests C values to balance **underfitting** (low C) & **overfitting** (high C).
- **Penalty**: **L1** removes useless features; **L2** shrinks weights toward zero.
- **Result**: selects the best parameter combination.

Parameter	Best
C	0.1
penalty	L2
solver	liblinear

- cv = 5: model trains on 4 and tests on 1, repeating for all folds.
- ie: **moderate regularization** (C = 0.1) with **L2 penalty** (shrinks weights) and the **liblinear** solver to give the best for small dataset.

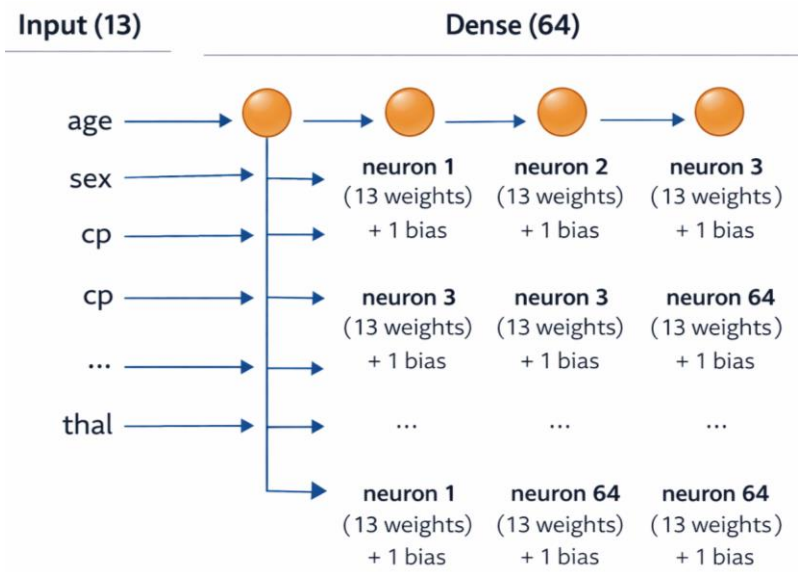
Model Development [LR] [2]

- **Top features**: cp (+0.60), thalach (+0.38), ca (−0.44)

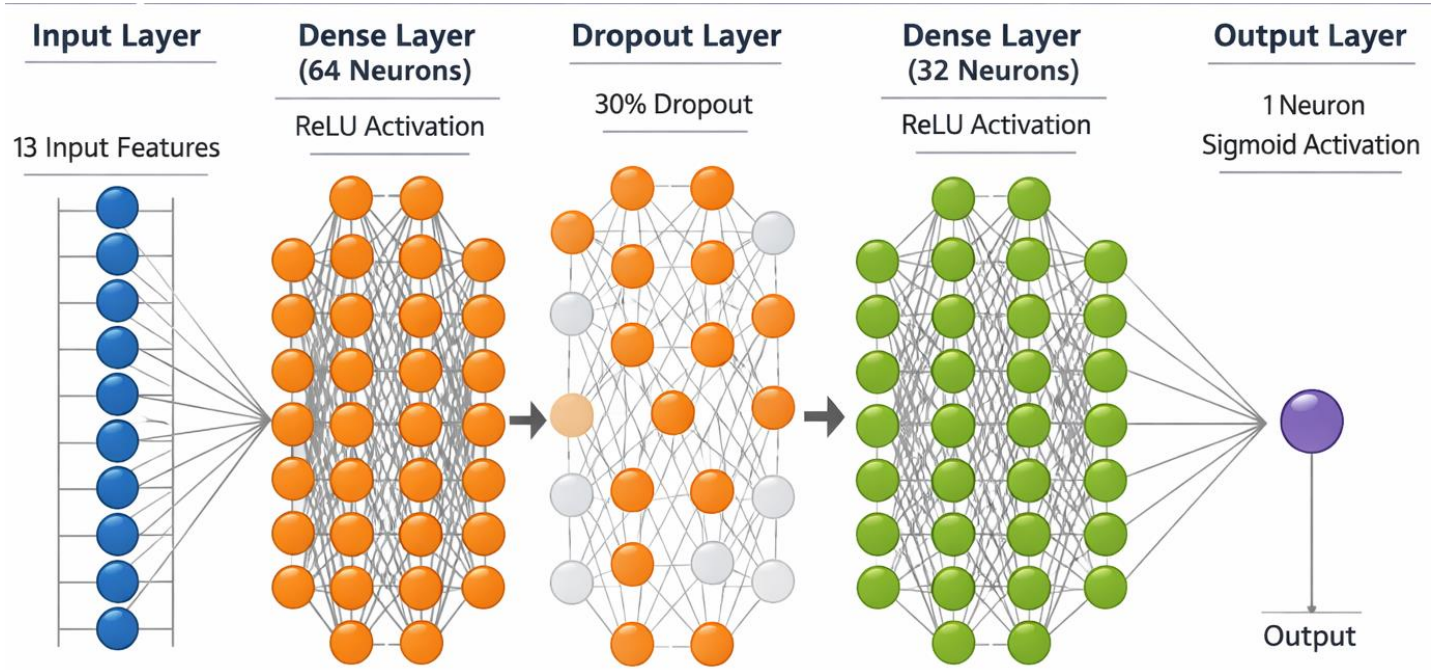


Model Development [ANN] [1]

- Each neuron: **13 weights** + **1 bias** = 14 parameters; 64 neurons >> **896** parameters, each learning different feature pattern

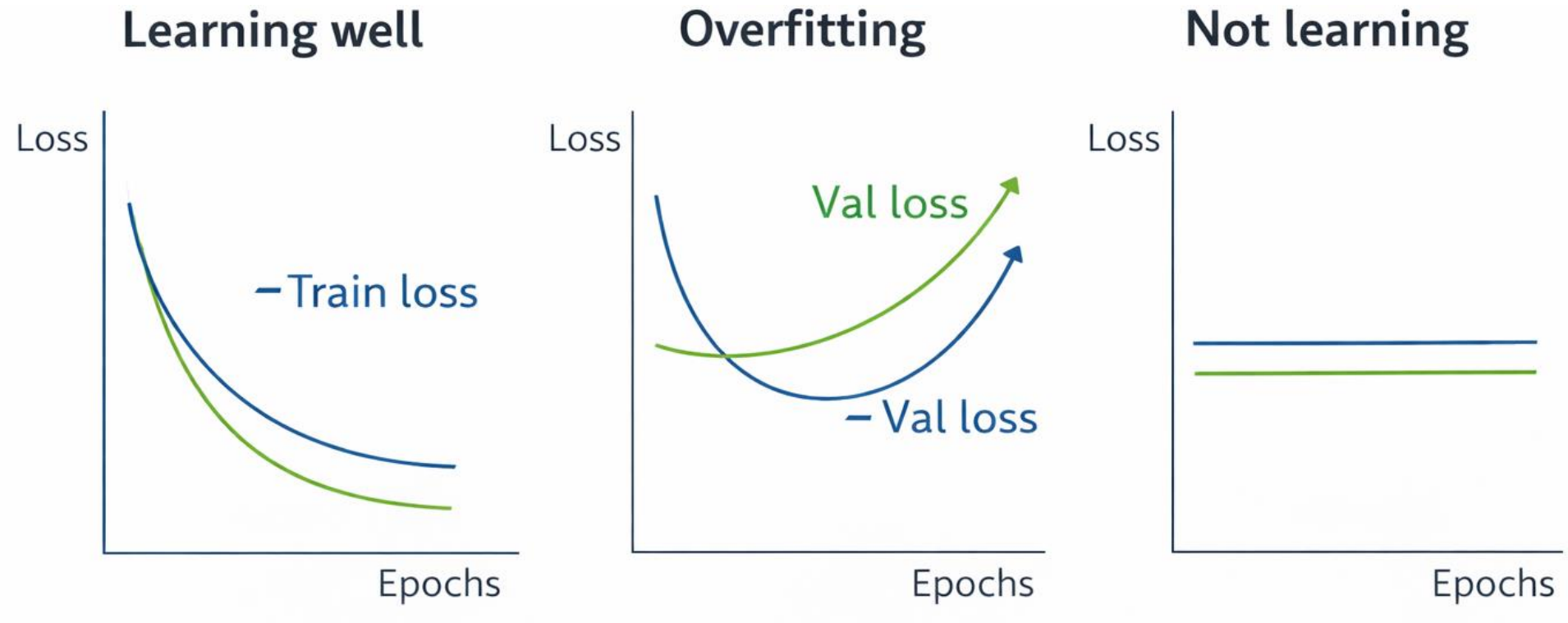


Dense: every input connects to every neuron.

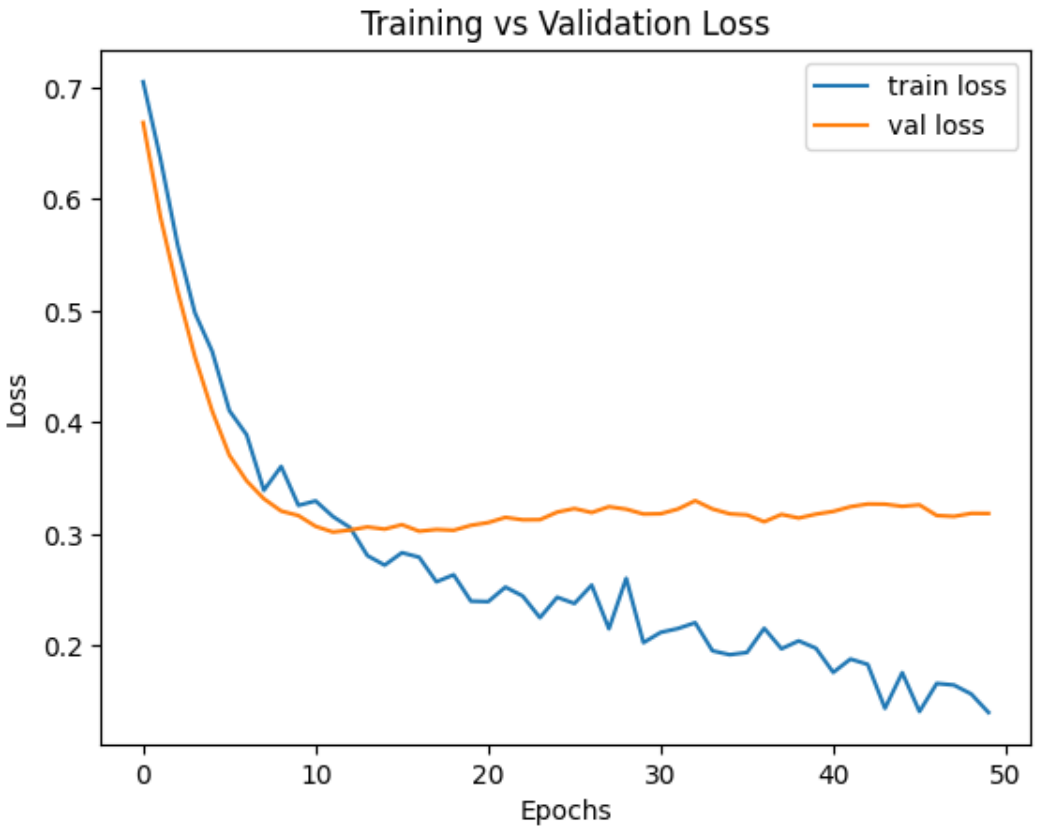


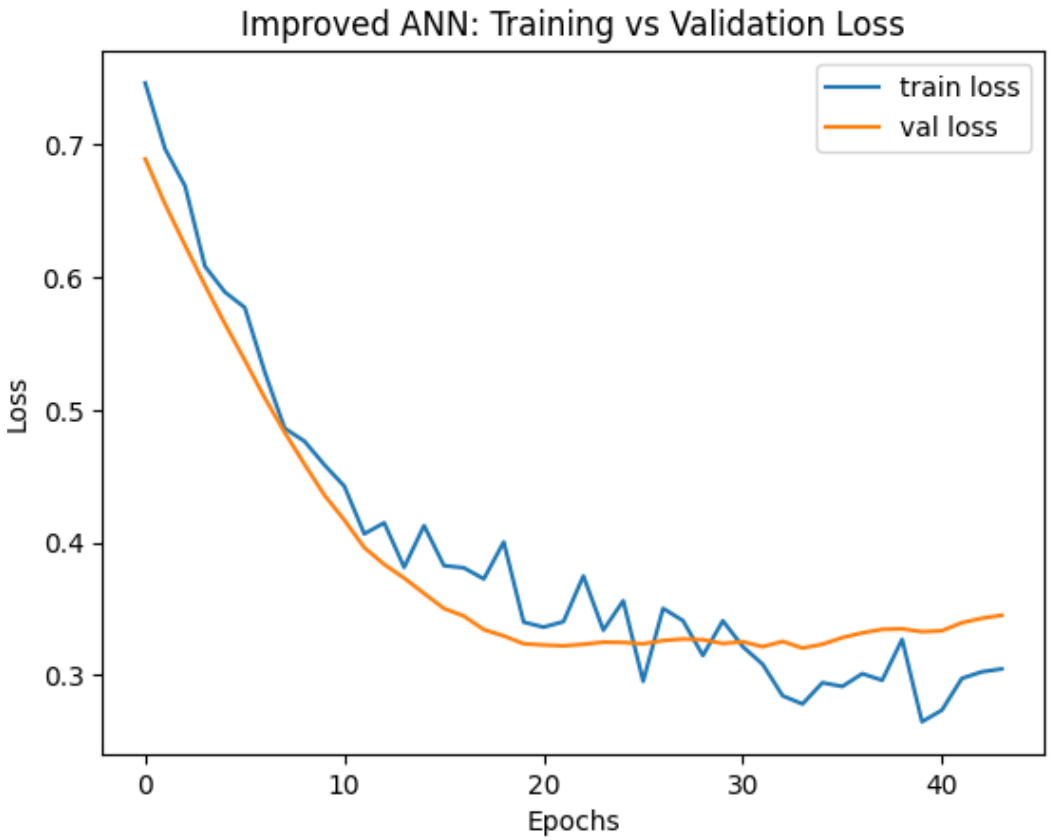
Model Development [ANN] [2]

- **Dropout**(0.3): ignores **30%** neurons → reduces **overfitting** (small dataset)
- **Optimizer**: **Adam** (adaptive learning rate) | **Loss**: Binary Cross-Entropy
- **Train loss**: error on training data | **Val loss**: error on validation data



Model Development [ANN] [2]





`64 -> 0.3 -> 32 -> 1 (all)`

`32 -> 0.4 -> 16 -> 1 (earlyStop)`

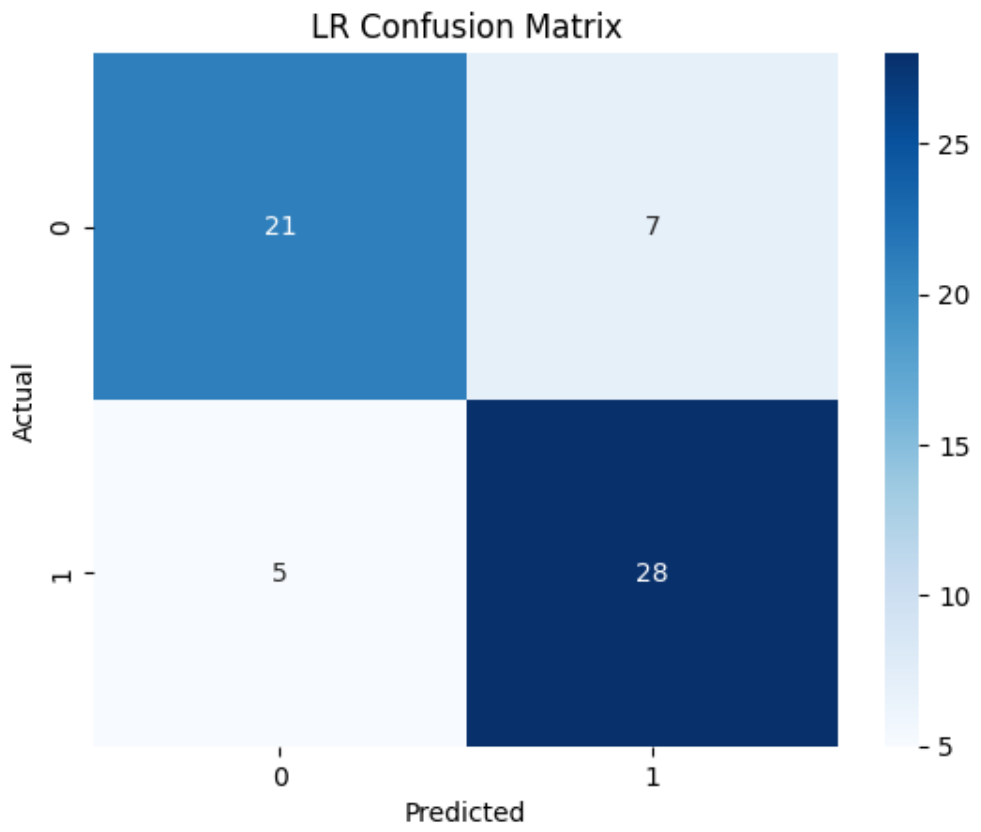
>> 50/50 split: initially learned, later overfitting.

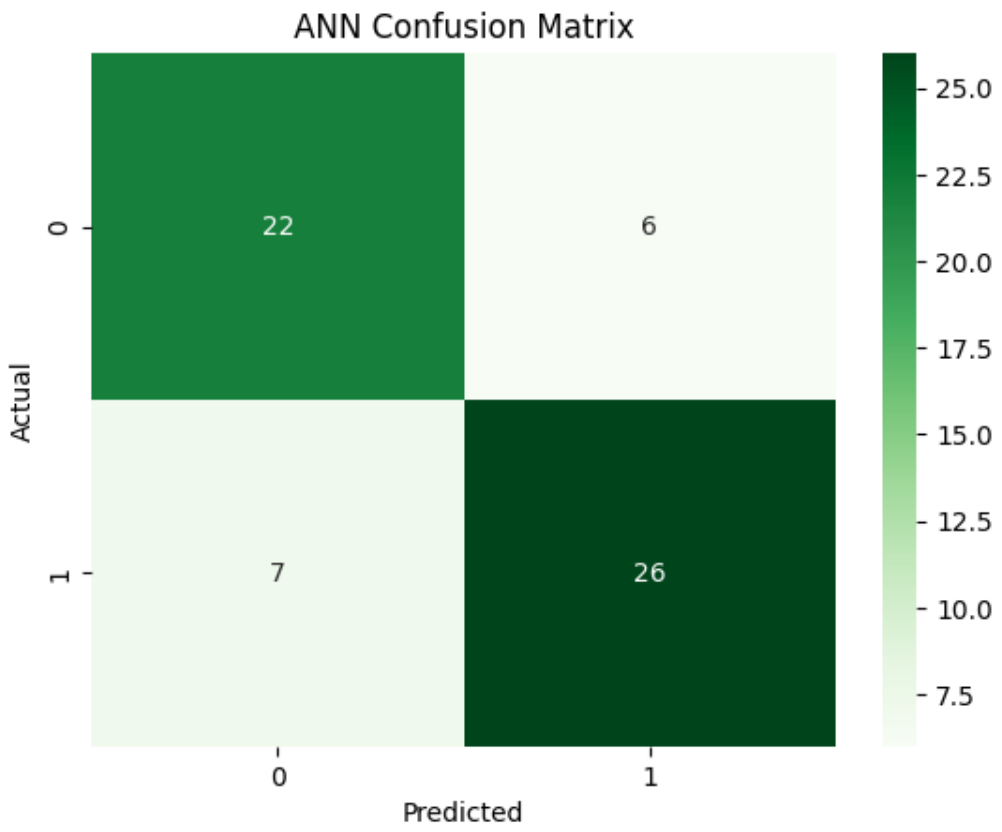
Metrics

Metric	Formula	Why it matters
Accuracy	correct / total	Overall correctness
Precision	TP / (TP + FP)	Fewer false alarms
Recall	TP / (TP + FN)	Catch sick patients
F1-Score	2PR / (P + R)	Balance of precision & recall

- **Precision** = of those **flagged** as disease, how many **actually had** it
- **Recall** = of **actual** disease cases, how many were **caught** ->
  most important in healthcare
  (**missing** a sick patient is **worse** than false alarm)

Confusion Matrices

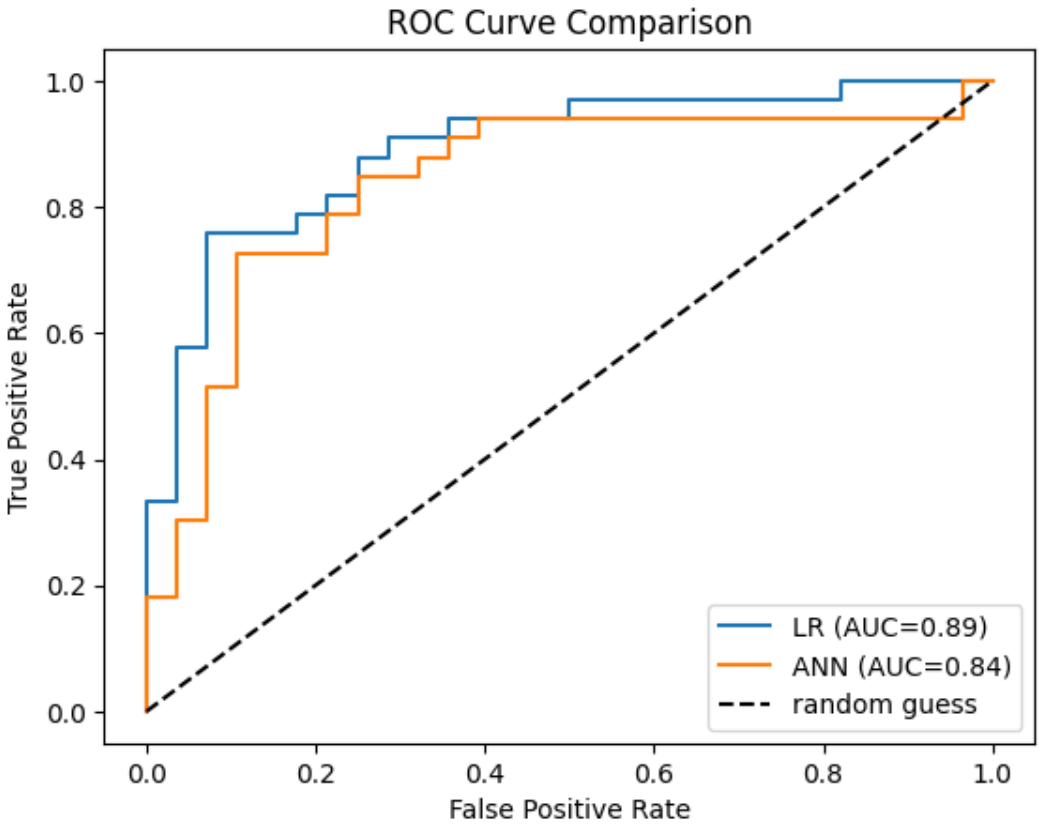




- Confusion matrices: **LR** caught **28/33** disease cases, **ANN** caught **26/33**
- LR missed fewer sick patients

ROC Curves & AUC

- **ROC** curve: **TPR** vs **FPR** across threshold (diagonal = random, AUC = 0.5)
- **AUC**: single summary (1 = perfect, 0.5 = random)
- **LR** = 0.86, **ANN** = 0.84: both good, LR slightly better



Results & Interpretation

Model	Accuracy	Precision	Recall	F1	AUC
LR	**0.803**	0.800	**0.848**	**0.824**	**0.892**
ANN	0.787	**0.813**	0.788	0.800	0.839

- **LR** wins 4/5 metrics
- **ANN**: slightly higher **precision** but misses more sick patients
- Reason: dataset small (302 samples); **ANNs** need more data
- **LR** advantage: interpretable coefficients → shows important features
- Top features: chest pain type, max heart rate, major vessels

Conclusion & Ethics

- **Conclusion**
  - Both models achieved ~80% accuracy
  - LR slightly beat ANN ; more complex ≠ always better
  - LR provides interpretability - coefficients reveal which features drive predictions

- **Ethics:**
  - Patient privacy important; dataset from one hospital >> may not generalize
  - Must never replace a doctor - assistive screening tool only
  - Future: larger datasets could favor ANN; ensemble methods (Random Forest) worth exploring

References

1. UCI ML Repository - Heart Disease Dataset
2. Kaggle - Heart Disease UCI Dataset
3. Krish Naik - Heart Disease Classification (YouTube)
4. StatQuest - Logistic Regression (YouTube)
5. 3Blue1Brown - Neural Networks (YouTube)
6. GitHub - huda-alq, kristipati, jhanvigoyal
7. scikit-learn, Keras, pandas, NumPy Docs

Tools: Python, pandas, NumPy, Matplotlib, Seaborn, scikit-learn, TensorFlow/Keras

Thank You